Hong Kong Institute of Vocational Education

Department of Information and Communications Technology

Final Year Project (ITP4913M)

Project Proposal

Please read the instructions in the Guidelines for Writing Project Proposal before completing this form.

Student Name(s):

Supervisor:

Project Title: Traffic and Pedestrian Peak Period Visualization System

Target Audience: general public

Statement of problems to be solved:

General: No a integrated platform which can visualizes the data for users to understand the city

Public: Drivers, Pedestrain (App version)

- Difficult to search for parking vanancy
- Difficult to know the crowded area
- Difficult to know the public transport information
- Difficult to know the biking route in the city
- Difficult to know whether there is congestion in some roads

Business: (App version & Web version)

- Difficult to know the area with high pedestrian flow
-

Government: (Web version)

- Not enough information for evaluation for future city plan
- Difficult to have information to declare the policy for improvement of the mobility

In the urban area, vehicle owners are difficult to find a place to park their car especially in the working hours since there is no real time information about the parking vacancy of the car parks nearby. They have no choice but to reach the car parks in order to inquire about the parking vacancy. Time is wasted if there is no vacancy available.

Hong Kong is one of the most densely populated cities in the world. Beyond looking for parking vacancy, overcrowding problem has been a common issue that people will experience when they go to some popular areas or shopping mall especially in the weekend. Since there is no information of pedestrian flow of specific areas, people have no idea where is less crowded.

At the weekends, many people like to relax themselves through biking. However, there is a lack of convenient ways to find the biking route.

Although there are apps which can provide information related to the public transport and biking route,it is inconvenient for the general users to seek and download a considerable numbers of apps to access the above information as there is no an app which centralizes it.

At the specific time, some roads may occur traffic jam since there are too many medium and large sized vehicles.One of the reason is that the authority cannot set up effective means to limit the number of medium and large sized cars when there is no information about the numbers of car type on the road..

Background to the problem:

Although information delivery has been swift thanks to the advancement of technology, it remains a major problem that people cannot obtain latest information related to their everyday life especially in topics of parking vacancy and pedestrian flow since there is no existing official platform which provides the above information.

Outline of proposed solution:

- Develop a Traffic and Pedestrian Peak Period Visualization System

Public: (Drivers, Pedestrain)

- Display the nearby/all car park to user (include details e.g parking vacancy, parking price)
- Display the nearby/all crowded area (include details e.g.Analysed Data)
- Display the nearby/all biking route (include details e.g Distance)
- Display the specific area of the public transport information (e.g various type of public transport's analys data)
- Display the degree of congestion on specific roads

Business:

- Display the areas with high pedestrain flow (include details e.g.Analysed Data)

Government:

- Provide analysed information(for further city development)

We will going to develop a Traffic and Pedestrian Peak Period Visualization System. To solve the difficulty of finding a parking vacancy of the car park, it is proposed that users can also clearly view the number of parking vacancy of the car parks nearby in the map with using GPS. Furthermore, Our apps can display car park detailed information e.g. address , opening hours and cost of the parking.

In terms of overcrowding problems, users can check the pedestrian flow of specific areas when they select the icon of the map. Pedestrian flow of specific areas will be displayed with certain standards such as low , normal and high so that user can easily know whether the area is crowded.

Concerning the traffic congestion at the specific time, our system can analyze the highest flow of the car types at which time going into these roads. And then some related department(Government) can plan initiatives to relieve the congestion problem with using the data such as restricting a specific type of car to enter specific road for a period of time.

Explanation of why the proposed solution is appropriate:

Public: (Drivers, Pedestrain)

- Problem : Difficult to search for parking vacancy
- Solution : Display the nearby/all car park to user (include details e.g Parking Space, Parking Price)
- Reason :There are no other tools on the market that can help.Our system can help vehicle owners to save time since they have no need to reach the car park and they can check whether there is any parking vacancy in the car park in advance when using our mobile application.

- Problem : Difficult to know the crowded area.
- Solution : Display the nearby/all crowded area (include details e.g.Analysed Data)
- Reason :If information of pedestrian flow of specific areas is provided ,certain proportion of people can go to the less crowded areas based on the the information. As a result, the overcrowding problem can be relieved and the users can change their route so that they can enjoy a more comfortable journey.

- Problem : Difficult to know the biking route in the city.
- Solution :Display the nearby/all biking route (include details e.g Distance)
- Reason : Our platform can let the public easily to know the biking route informations.

- Problem : Difficult to know whether there is congestion in some roads
- Solution :Display the degree of congestion on specific roads
- Reason : Our system can show the degree of congestion on specific roads based on the road condition

Government:

- Problem : Difficult to know the public transport information.
- Solution : Display the specific area of the public transport information (e.g various type of public transport's analyzed data).
- Reason : Providing the useful public transport information for the public including the Taxi, Bus , MTR and Mini Bus.

- Problem : Difficult to know the crowded area.
- Solution : Display the nearby/all crowded area (include details e.g.Analysed Data)
- Reason : Information can be used by the police and inspectors to monitoring the pedestrian flow which suddenly grows up in order to prevent accident ,illegal assembly. Also, it will be useful to count the number of people in a parade

Business:

- Problem : Difficult to know the high pedestrian flow area
- Solution : Display the flow of high pedestrian flow area (include details e.g.Analysed Data)
- Reason : Providing the useful information and use to develop a marketing strategy. For example, estate agency company can set a more accurate price of the property based on the pedestrian flow. On the other hand, people who are looking for premises can make use of the pedestrian flow data to determine whether the premise is located in a street with frequent high pedestrian flow.

Main stages involved in the project, with estimated time to completion of each stage:

1. analyze the data of parking vacancy provided by the government and visualize them in the map. Estimated time completion is 1 month
2. recognize the pedestrian and vehicle flow with using AI technogy. Estimated time completion is 1 to 3 months
3. Collecting the data about pedestrian and vehicle flow, and then analyzed by the system. Estimated time completion is 1 months
4. Develop programs for counting the pedestrians and vehicles. Estimated time completion is 3 months
5. Design the user interface of the mobile apps and the implement the functions of the apps. Estimated time completion is 2 months
6. Develop a web system that contains the detailed processed data and represents the data with graphs. Estimated time completion is 1 months
7. Convert the processed data into json format and allow the data can be accessed by the public. Estimated time completion is 1 to 2 weeks
8. Implement functions of providing public transport as well as biking route. Estimated time completion is 1 to 2 weeks

Main Deliverables:

An apps that can enable public users to check the parking vacancy , pedestrian flow and vehicle flow. It can also offer public transport as well as biking route

A web system that offers analyzed data which derives from parking vacancy , pedestrian flow and vehicle flow. The data will be represented with not only numbers but also graphs.

The (tentative) responsibilities of each member:

Student A : study map and web and related API

Student B : study programs related to AI and API

Student C : Study MAP API, AI and Web

Student D : study programs related to AI and Android

Projected Resource Requirements:

Hardware (Computer/any specialised hardware):
Carmera (e.g CCTV)
Computer
Server
Mobile Phone

Software:
Anaconda
OpenCV API
Tensorflow
Cloud Server
Google Map API

Owner and location of resources:
Alibaba, google,

External involvement:

Advice given by Supervisor:

Advice given by Second Reader:

Student signature: Date:

Supervisor signature: Date:

Second Reader signature: Date:

External Body signature (if relevant): Date: